

— LYZR AGENTICOS

Agentic AI for Supply Chain *& Logistics*

Twelve agents · One platform · First deploy in six weeks

Supply chains are losing billions *to manual planning* and siloed systems.

Planners spend most of the week firefighting exceptions. Dashboards show what broke — rarely what to do. Legacy planning engines deploy in years, not weeks. The outcome: millions in lost revenue and eroded margins every quarter.

60–70%

Planner time spent on manual data work

\$184B

Global trade-friction cost annually

8–12%

EBITDA exposed to volatility

12%

Enterprises with fully automated planning

From dashboards *to autonomous execution.*

Enterprises move through five autonomy tiers. Most are stuck at tiers 1–2. Agentic AI is what makes tiers 3–5 economically viable for the first time.

01

Observe

Dashboards show what is happening.

02

Suggest

System flags anomalies and trends.

03

Recommend

Context-rich actions proposed to humans.

04

Auto-execute

Agent acts under human approval.

05

Full auto

Closed-loop execution with guardrails.

Twelve agents *across the value chain.*

Plan · Source · Make · Deliver & Protect — a unified portfolio that composes cross-module workflows through a shared event bus and decision ledger.

01

Plan

Forecast · balance · right-size stock.

- Demand Sensing & Forecasting
- Supply–Demand S&OP
- Inventory Optimization

3 AGENTS

02

Source

Buy smarter · de-risk suppliers.

- Spend Intelligence & Savings
- Supplier Risk & Resilience

2 AGENTS

03

Make

Produce reliably · quality built-in.

- Production Scheduling (APS)
- Predictive Maintenance & Quality

2 AGENTS

04

Deliver & Protect

Promise · move · recover · comply.

- Order Promising & ATP
- Logistics Route Optimization
- Warehouse Task Orchestration
- Returns & Reverse Logistics
- Trade Compliance & Documentation

5 AGENTS

The enterprise *outcome stack*.

Across the twelve-agent portfolio, industry research consistently supports the following outcome bands for AI-led supply-chain operating models.

≤12%

Forecast MAPE achievable

95%+

On-time, in-full (OTIF)

15–25%

Working capital freed

4–8%

Procurement savings

3–6 MO PAYBACK ON FIRST AGENT

5–8× ROI IN YEAR ONE

8–12% EBITDA UPLIFT AT FULL PORTFOLIO

20–50% PLANNER TIME FREED

01 DEMAND SENSING & FORECASTING

Know demand *before it happens.*

Ingests real-time POS, market, weather, and social signals to generate hyper-accurate short and mid-term demand forecasts — continuously learning and self-correcting.

BUILT FOR DEMAND PLANNERS · SUPPLY CHAIN VPS · S&OP LEADERS · CIOs

30–50%

Forecast accuracy uplift

60%

Manual planning time cut

20–35%

Excess inventory reduced

85–95%

SKU-level accuracy

01

Data ingestion

POS, ERP, weather, social and other signals.

02

Signal processing

AI normalises and correlates multi-source inputs.

03

ML forecasting

ARIMA, Prophet, LSTM, Transformer ensembles.

04

Anomaly detection

Flags demand spikes and trend shifts.

05

Action & alerts

Auto-triggers replenishment and notifies planners.

02 SUPPLY-DEMAND BALANCING & S&OP

Align every function *in real time.*

Continuously aligns supply, demand, finance, and operations on real-time data — compressing planning cycles from weeks to hours and enabling autonomous cross-functional decisions.

BUILT FOR S&OP LEADERS · SUPPLY CHAIN VPS · CFOS / FP&A · COOS

25–40%

Forecast bias reduction

70–80%

Planning cycle time cut

30–70%

Planning FTE cost reduction

90–95%

Demand-plan accuracy

01

Signal ingestion

Aggregates demand, supply, finance, market signals.



02

Cross-functional alignment

Sales, Ops, Finance & Supply on one model.



03

Concurrent planning

Simultaneous scenarios across constraints.



04

Exception management

Flags imbalances and triggers risks in real time.



05

Decision execution

Approves plans, updates ERP, alerts stakeholders.

03 INVENTORY OPTIMIZATION

Right stock. Right place. *Right time.*

Autonomously monitors stock, supplier performance, and demand signals — triggering replenishment, reallocating inventory, and preventing disruptions before they occur.

BUILT FOR INVENTORY MANAGERS · SC DIRECTORS · DEMAND PLANNERS · COOS

20–30%

Carrying cost reduction

35–45%

Stockout events cut

15–25%

Working capital freed

60%

Manual replenishment work freed

01

Inventory monitoring

Real-time stock visibility across all nodes.



02

Demand sensing

AI reads trends, seasonality and external signals.



03

Replenishment planning

Optimal order quantity, timing and supplier mix.



04

Disruption response

Detects stockout risk and triggers corrections.



05

Autonomous execution

Places POs, reallocates stock, updates ERP.

04 PRODUCTION SCHEDULING (APS)

Schedules *that adapt instantly.*

Continuously optimises job sequences, machine assignments, and shift plans using real-time shop-floor data — adapting to disruptions and maximising throughput on every line.

BUILT FOR PRODUCTION PLANNERS · PLANT MANAGERS · OPERATIONS DIRECTORS · CIOs

20–30%

OEE improvement

15–25%

Setup time reduction

90–100%

Schedule adherence

20–40%

Planner hours saved per week

01

Shop-floor data sync

IoT, MES, ERP and sensor streams in real time.



02

Constraint analysis

Maps capacity, tools, skills, materials.



03

Schedule optimization

AI sequences jobs to minimise setup, maximise OEE.



04

Disruption adaptation

Re-sequences on breakdowns, rush orders, delays.



05

Plan execution

Pushes schedules to MES, ERP and the shop floor.

05 SPEND INTELLIGENCE & SAVINGS

Find savings *hiding in plain sight.*

Ingests, cleanses and classifies 100% of spend, identifies consolidation and savings opportunities, and generates ranked sourcing strategies — typically uncovering 8–15% addressable savings within 90 days.

BUILT FOR CPOS · PROCUREMENT MANAGERS · CATEGORY MANAGERS · CFOS

8–15%

Addressable savings uncovered

30–50%

Maverick spend leakage cut

100%

Spend classified & taxonomised

60–80%

Analysis time reduction

01

Spend ingestion

Invoices, POs, card data from all P2P systems.



02

Cleanse & classify

AI maps 100% of spend to UNSPSC / custom taxonomy.



03

Opportunity ID

Finds consolidation, maverick and price variance.



04

Supplier analysis

Benchmarks pricing, share and category trends.



05

Savings recommendation

Ranked strategies with projected ROI.

06 SUPPLIER RISK & RESILIENCE

See risk *before it hits.*

Ingests real-time signals from news, financials, geopolitics and performance data to continuously score every supplier — detecting risks 40–60% faster and auto-triggering mitigation before disruption.

BUILT FOR CPOS · RISK MANAGERS · SC DIRECTORS · COOS

40–60%

Faster risk detection

30–50%

Supply disruption reduction

3–5×

Supplier risk coverage

20–35%

Sourcing risk cost reduction

01

Risk signal ingestion

News, financials, weather, sanctions, ESG feeds.

02

Supplier scoring

AI scores every supplier across 50+ dimensions.

03

Risk classification

Categorises financial, geopolitical, ESG, ops.

04

Alert & escalation

Context-rich alerts routed to right stakeholders.

05

Mitigation planning

Recommends backup suppliers and rebalance options.

07 ORDER PROMISING & ATP

Promise what *you can deliver.*

Dynamically checks inventory ATP and production capacity across all nodes in real time — achieving 95–99% promise accuracy and eliminating costly fulfilment failures.

BUILT FOR ORDER MANAGEMENT · SUPPLY PLANNERS · SALES OPS · COOS

95–99%

Order promise accuracy

30–50%

Fulfilment failure reduction

15–25%

Revenue recovered from lost orders

20–30%

Expedite cost reduction

01

Order intake

Orders captured from every channel real-time.

02

Inventory ATP check

Real-time availability across DCs, stores, transit.

03

Capacity ATP check

Validates production, labour, machine constraints.

04

Dynamic promising

AI calculates optimal delivery date with confidence.

05

Order confirmation

Commits delivery and triggers fulfilment.

08 LOGISTICS ROUTE OPTIMIZATION

Every route. *Optimized.*

Autonomously selects carriers, optimises routes in real time, monitors every shipment, and resolves exceptions — cutting freight cost 15–25% while achieving 95%+ on-time delivery.

BUILT FOR LOGISTICS MANAGERS · TRANSPORTATION MANAGERS · OPERATIONS VPS

15–25%

Freight cost reduction

30–40%

Transit time variability cut

95%+

On-time delivery achieved

10–20%

Carbon footprint reduction

01

Shipment intake

Order & shipment data from all channels.



02

Carrier selection

AI scores carriers on cost, reliability and SLA.



03

Route optimization

Multi-stop routing with live traffic and weather.



04

Real-time tracking

End-to-end visibility with predictive ETA.



05

Exception management

Auto-reroutes or alerts stakeholders on delays.

09 WAREHOUSE TASK ORCHESTRATION

Orchestrate *every movement.*

Dynamically prioritises orders, assigns tasks to workers and robots, and optimises pick paths in real time — lifting throughput 20–30% while cutting labour cost 15–25%.

BUILT FOR WAREHOUSE MANAGERS · DC OPS LEADERS · LOGISTICS DIRECTORS · COOS

20–30%

Warehouse throughput gain

15–25%

Labour cost reduction

25–40%

Pick-path efficiency gain

99%+

Order accuracy achieved

01

Order prioritization

AI ranks orders by SLA, cut-off and urgency.



02

Task assignment

Dynamically assigns to workers, AMRs, conveyors.



03

Pick-path optimization

Calculates optimal routes; cuts congestion.



04

Exception handling

Detects shorts, damage, mis-picks; reassigns.



05

Shipment execution

Consolidates orders, prints labels, triggers dock.

10 PREDICTIVE MAINTENANCE & QUALITY

Fix it *before it breaks.*

Continuously monitors IoT/SCADA signals, detects anomalies, predicts failures, and autonomously schedules maintenance — eliminating unplanned downtime and catching defects before they leave the line.

BUILT FOR MAINTENANCE MANAGERS · PLANT MANAGERS · QUALITY DIRECTORS

20–40%

Unplanned downtime reduction

25–35%

OEE improvement

40–60%

Quality defect reduction

15–30%

Maintenance cost reduction

01

IoT data ingestion

SCADA, MES and sensor streams continuously.



02

Anomaly detection

AI identifies abnormal patterns across equipment.



03

Failure prediction

ML models predict failure timing and root cause.



04

Maintenance scheduling

Auto-schedules work orders optimally.



05

Quality inspection

Vision AI detects defects and triggers holds.

11 RETURNS & REVERSE LOGISTICS

Turn returns *into recovery*.

Automates return authorisation, AI-grades items, makes instant disposition decisions, and routes each item to its highest-value recovery channel — recovering 15–25% more value.

BUILT FOR REVERSE LOGISTICS MANAGERS · RETAIL SC LEADERS · E-COMMERCE OPS

30–50%

Returns processing time cut

15–25%

Recovery value uplift

40–60%

Disposition decision time cut

20–35%

Refurbishment yield increase

01

Return request

Auto-validates against policy rules.



02

Inspection & grading

AI vision grades A/B/C/D with defect detail.



03

Disposition decision

Optimal channel: restock, refurbish, donate, scrap.



04

Routing & fulfilment

Routes item to best recovery site.



05

Credit processing

Auto-issues credit and updates ERP financials.

12 TRADE COMPLIANCE & DOCUMENTATION

Compliance *without the overhead.*

Autonomously classifies products, generates compliant trade docs, validates against current regulations, and submits to customs — reducing doc time 70–85% and compliance errors 90–95%.

BUILT FOR TRADE COMPLIANCE · LOGISTICS DIRECTORS · CUSTOMS TEAMS · CFOS

70–85%

Documentation time cut

90–95%

Compliance error reduction

40–60%

Customs clearance time cut

30–50%

Customs penalty risk reduction

01

Shipment intake

Order & product data captured from ERP / OMS.



02

Regulatory classification

AI assigns HTS / HS codes and duty rates.



03

Document generation

Auto-creates COO, invoices, packing lists, SLI.



04

Compliance validation

Cross-validates against sanctions and trade rules.



05

Customs submission

Submits to ACE, CDS and partner portals electronically.

Four personas. *One shared operating picture.*

Each persona gets agent outputs in their tool of choice — no new UI to learn. Cross-persona workflows stitch through the unified event bus.

HERO PERSONA

CSCO

Weekly executive brief. Scenario portfolio. OTIF, working-capital and resilience index.

USES

S&OP • INVENTORY • LOGISTICS • SUPPLIER RISK

HERO PERSONA

VP Demand Planning

Consensus forecast workbench. Exception queue. MAPE and bias tracking.

USES

DEMAND SENSING • S&OP • INVENTORY

HERO PERSONA

CPO

Category P&L. Risk radar. Savings pipeline and sourcing tower.

USES

SPEND INTELLIGENCE • SUPPLIER RISK • TRADE COMPLIANCE

HERO PERSONA

S&OP / Inventory Head

Daily cockpit. Inventory health. E&O heat-map. Action queue.

USES

S&OP • INVENTORY • ATP • LOGISTICS

Value starts *in weeks*, not years.

A typical agentic deployment moves through four stages — each delivering measurable outcomes before the next one starts.



Built to act, *not just analyse.*

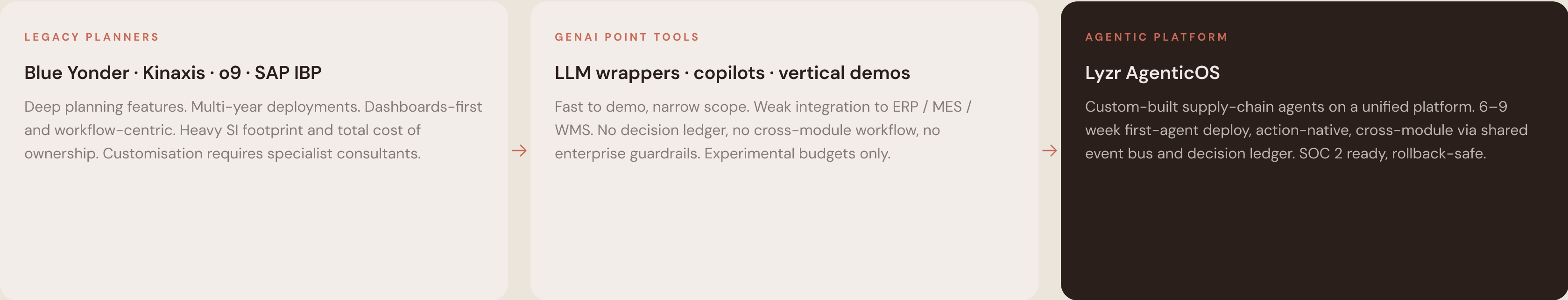
Signals → decision → agents → actions → governance. Every agent plugs into a unified event bus and decision ledger — enterprise-grade from day one.



- ONE PLATFORM
- TWELVE AGENTS
- ONE DECISION LEDGER
- EACH NEW AGENT INHERITS THE FULL INTEGRATION SURFACE

Three approaches *enterprises buy today.*

Each category delivers different trade-offs across depth, speed, integration surface and autonomy. The right choice depends on what the buyer is trying to un-stick.



Lyzr vs. *the alternatives.*

Eight buying dimensions. Each dot = 1 point (5 = best-in-class). Scores reflect public vendor documentation and analyst coverage.

DIMENSION	LYZR AGENTICOS	BLUE YONDER	KINAXIS	O9 SOLUTIONS	PALANTIR FOUNDRY
Pre-built SC agents					
Autonomy model					
Agent-native architecture					
Time-to-first-agent					
Multi-ERP integration					
Decision audit trail					
Human-in-the-loop & rollback					
TCO shape					

Start small. *Scale fast.*

Speed to first value determines the full-portfolio arc. Agent infrastructure collapses the land phase to weeks — and each subsequent agent compounds.

6 *weeks*

First agent live in production

6–9 *mo*

Payback on the first agent

5–8×

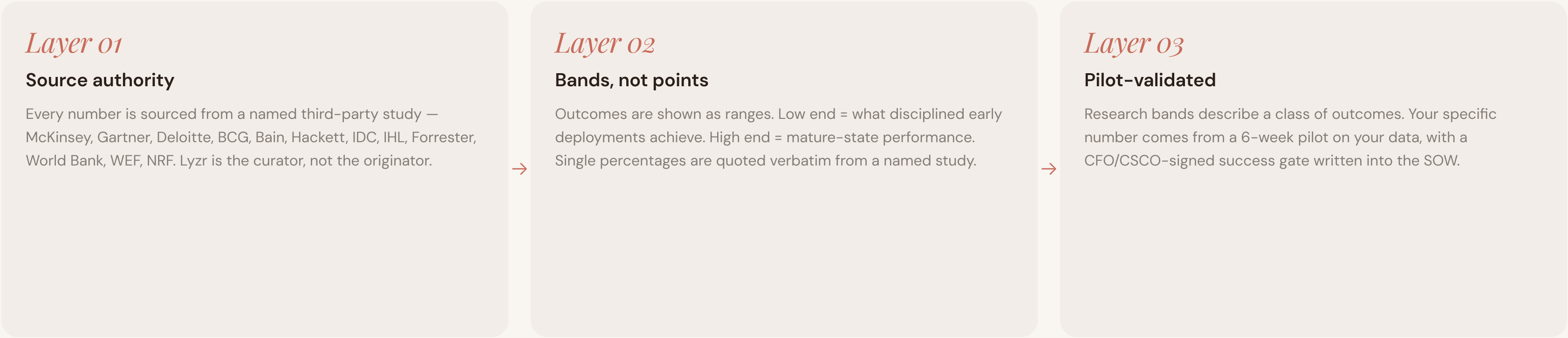
ROI in year one

8–12%

EBITDA uplift at full portfolio

How to read *every number* in this deck.

Three independent layers back every KPI, ROI, and outcome band on the slides. Any one answers 80% of the objections on its own; together they remove the ambiguity.



Why ranges — *and how* they are constructed.

We avoid single-point claims because supply chains differ. The band captures the distribution of observed outcomes across operating models, data maturity, and scope — exactly what a responsible buyer should expect to see.

Rule A

Union range

Where 2+ studies cover the same KPI, we report the union — lowest low to highest high. Narrower sub-bands cited on the slide for clarity.



Rule B

Single source

Where only one credible source exists, we cite it by name on the slide. No interpolation; the provenance is visible.



Rule C

No double-counting

Portfolio-level numbers (e.g., 8–12% EBITDA uplift) are drawn from studies that measured portfolio outcomes directly — never summed from per-agent figures.



Rule D

Position within band

Baseline, data maturity, process readiness and scope determine where you land. Narrow-scope pilots land near the low end; cross-module at maturity reaches the high end.

How the research band *becomes your number.*

A 6-week pilot on your data, under a signed SOW. Baseline measured against your current state. Gate signed by your CFO and CSCO. We do not scale without hitting the gate.

Week 1

Baseline

Current-state KPIs on your actual data — MAPE, OTIF, inventory days, planner hours. Your number, not an assumed one.

Week 1–2

Signed gate

Co-defined pass/fail. Numeric improvement on a pre-selected KPI, measured at week 6, with statistical significance threshold written in.

Weeks 2–5

Shadow & measure

Agent runs in tier-1 autonomy alongside planners. Every recommendation scored against what they did. Tier escalation only on your authorisation.

Week 6

Gate decision

Measured vs. baseline and gate. Hit the gate → we scale. Miss the gate → we don't. The research band is informational; the gate is contractual.

What Lyzr commits to — *and what it does not.*

Honest framing up-front removes the biggest stall in procurement. Platform commitments sit in the MSA; outcome commitments sit in the pilot SOW; research bands stay informational.

LYZR COMMITS TO · CONTRACTUAL

In writing

- Platform uptime, response-time SLA, integration milestones — in the MSA.
- Pilot success gate with financial consequences if missed — in the SOW.
- SOC 2 Type II, RBAC, audit ledger, BYOK, private cloud — standard.
- Customer owns integration IP and pilot baseline data — default.
- Rollback-safe architecture — every agent write is reversible.

LYZR DOES NOT · INFORMATIONAL

Not contractual

- Platform-level %-improvement guarantees — outcomes depend on your data and scope. No industry vendor SLAs these.
- Research-band figures as guarantees — bands are industry-observed, not Lyzr-promised.
- Year-1 ROI on a fixed timeline beyond pilot scope — depends on agents rolled out.

Sources *& methodology.*

MCKINSEY & COMPANY

Succeeding in the AI supply-chain revolution (2021) · Rewiring operations for resilience (2023) · The state of AI (2024) · The future of maintenance (2022) · Logistics 2030 (2023).

GARTNER

Supply Chain Planning Maturity Survey (2023) · Supply Chain Top 25 Methodology · Magic Quadrant for SCP Solutions (2023) · WMS Market Guide (2023) · Order Management Research (2023) · Hype Cycle for Supply Chain Planning (2023).

DELOITTE

Global CSCO Survey (2023) · The CSCO Agenda (2024) · CPO Survey (2023) · Smart Factory Study (2022) · Supply Chain Digital Transformation Study (2023) · Predictive Maintenance Benchmarks (2023) · The Future of Supply Chain (2023).

BAIN & COMPANY

Working Capital Excellence (2023) · Procurement Excellence Benchmarks (2023) · The Agile Supply Chain (2023).

BCG

Digital Acceleration in Supply Chain (2023).

THE HACKETT GROUP

Digital Procurement Study (2023).

IDC

Worldwide AI-Powered Supply Chain Forecast (2023) · MarketScape Supply Chain Planning (2023) · Reverse Logistics Market Forecast (2023) · Worldwide Transportation Management (2023).

OTHER

IHL Group Warehouse Robotics Study (2023) · Forrester Wave: Supply Chain Planning (2023) · World Bank Trading Across Borders (2023) · WEF Global Enabling Trade Report · NRF Retail Returns Study (2023).

METHODOLOGY

Outcome ranges represent industry-observed bands from the cited research. Per-agent slides list the primary sources feeding those metrics. Composite figures (slide 5) are drawn from the union of cited studies, not additively combined.

— CLOSING

The supply-chain moment
belongs to systems *that act.*

Build your first agent.